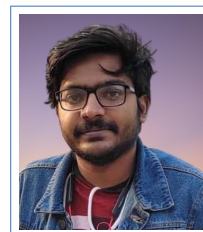


Sanchayan Banerjee

Curriculum Vitae

School of Physical Sciences
National Institute of Science Education and Research
☎* +91 7076453352
✉[regular] sanchayan.banerjee@niser.ac.in



Education

- Aug 2021 – Present **Integrated MSc-PhD, Physical Sciences**, National Institute of Science Education and Research, Bhubaneswar, (5th year Integrated MSc-PhD student)
CGPA: 9.03 and SGPA: 8.4 (1st sem), 8.6 (2ndsem), 8.2 (3rd sem), 9.0 (4th sem), 10.0(5th sem), 10.0(6th sem)
- (2018 – 2021) **Undergraduate (B.Sc. physics)**, Department of Physics, Ramakrishna Mission Residential College : (Narendrapur), University of Calcutta., CGPA : 8.55 / 10
- (2018) : **Grade XII (10+2 Class)**, Dwarhatta Rajeshwari institution, Hooghly, West Bengal Council of Higher Secondary Education, Total : 96.4 Percent (Marks), PHYSICS (96/100), MATH (100/100)
- (2016) : **Grade X (10 Class)**, Dwarhatta Rajeshwari institution, Hooghly, West Bengal Council of Higher Secondary Education, Total : 91.4 Percent (Marks)

Research Interest

- Many Body Localization Many-body localization phase transitions, Interplay of U(1) gauge fields and disorders, Breakdown of Eigenstate Thermalization Hypothesis (ETH): Study of Hilbert Space Fragmentation, Quantum Many-Body Scar using quantum dynamics (OTOC, Entanglement Entropy), Breakdown of ETH in interacting periodically driven and non-Hermitian systems.
- Non-Hermitian and driven systems Dynamics of Interacting Non-Hermitian systems using Exact Diagonalization and TDVP(Itenor). Floquet engineering and reentrant phases in disordered systems.
- Many Body Ground State Properties Ground state properties of strongly correlated systems. Study of interacting symmetry-protected topological phases using DMRG and relaxation dynamics with Exact Diagonalization and TDVP. Frustrated systems, as well as higher-dimensional systems using neural network techniques and quantum circuits.
- QFI Quantum Fisher Information and its applications in quantum sensing

Computational Skills

- Languages & tools **Python and C++: Exact Diagonalization, Quspin, Qutip, Density Matrix Renormalization Group (DMRG) (using TENPY and Itensors library), Time Dependent Variational Principle (TDVP) of MPS based on ITensors, Neural networks (netket), Amazon Braket Quantum Computing Module**
- Operating systems Windows, Linux, MacOS X.
- Applications LaTeX, gnuplot, Mathematica.

Publications

- 2024 **Emergence of distinct exact mobility edges in a quasiperiodic chain**, Sanchayan Banerjee, Soumya Ranjan Padhi, Tapan Mishra, *Physical Review B Letters*.
- 2024 **Reentrant topology and reverse pumping in a quasiperiodic flux ladder**, Sanchayan Banerjee, Rajashri Parida, Tapan Mishra, Preprint. [arXiv:2508.04452](https://arxiv.org/abs/2508.04452)
- 2024 **Multiple many-body localization transitions in a driven non-Hermitian quasiperiodic chain**, Sanchayan Banerjee, Ayan Banerjee, Tapan Mishra, Flore K. Kunst, Preprint. [arXiv:2509.14696](https://arxiv.org/abs/2509.14696)

- 2024 **Emergence of non-trivial phases in interacting non-Hermitian quasiperiodic chains with power-law hopping**, Aditi Chakrabarty, Sanchayan Banerjee, Tapan Mishra, Sanjoy Datta, Preprint. [arXiv:2508.14724](#)
- 2024 **Anomalous spectrum in a non-Hermitian quasiperiodic chain**, Soumya Ranjan Padhi, Sanchayan Banerjee, Tanay Nag, Tapan Mishra, *Physical Review B*.
- 2024 **Quasiperiodic and periodic extended Hatano–Nelson model: Anomalous localization transition and non-Hermitian skin effect**, Soumya Ranjan Padhi, Ashirbad Padhan, Sanchayan Banerjee, Tapan Mishra, *Physical Review B*.
- 2024 **Non-Hermitian comb effect in coupled clean and quasiperiodic chains**, Soumya Ranjan Padhi, Souvik Roy, Biswajit Paul, Sanchayan Banerjee, Tapan Mishra, Preprint. [arXiv:2509.26615](#)

Experience

- Jan 2023 - Jun 2024 MSc Thesis Defence: Disorder and Flux Induced Reentrance Topological Phase transition and Thouless pumping.
- Dec 2022 Arranged a Workshop for young researchers where senior research scholars from all over the world delivered talks on various areas of physics. Link of YPM 2022
- April – July 2022 Done a Research project under Prof. Satyendra Nath Gupta from PRL, Indian Space Research Organization (ISRO), India on Enhancement of Quantum efficiency of Quantum Emitters.
- Nov - Dec 2021 Done a Project on Spintronics before Prof. Satyaprasad P Senanayak, (Assistant Professor, Physical Sciences, NISER) and Prof. Shovon Pal (Assistant Professor, Physical Sciences, NISER)
- Aug 2020 – Aug 2021 Completed a B.Sc. Thesis project named “ The motion of a Space-ship in a rotating Space-time (Blackhole)” under Prof. Srijit Bhattacharjee , Assistant Professor, Indian Institutes of Information Technology (IIIT) Allahabad, India.
- July - Dec 2022 Completed Semester project under Prof. Luke Robert Chamandy (Prof. of NISER, previously from University of Rochester) on “On the energy and formalism of the Common Envelope Evolution: Open questions and Questionable Openings” (Process of merging of Neutron Stars).

School and Conferences attended

- 2025 Attended **Computational Methods in Lattice Field Theories** in IISC, Bangalore.
- 2025 Attended **Physics of Quantum Matter School(PQMS-2025)** in IISER Pune.
- 2024 Attended **School on Classical and Quantum Long-range Interacting Systems during 23 September 2024 – 4 October 2024** in TIFR, Mumbai.
- 2024 Attended **Topical School of Advanced Condensed Matter Physics** in IOP, Bhubaneswar.
- 2023 Attended **Physics of Quantum Matter School(PQMS-2023)** in NISER, Bhubaneswar.
- 2022 **Summer School on Quantum Information and Quantum Technology From IQIT 2022, IISER Kolkata.**
- 2022 **Amazon Braket Workshop on Quantum Computation:** Topics :: evaluating quantum circuits on quantum simulators and quantum hardware, simulating the effects of noise, use of the hybrid jobs feature, quantum machine learning.
- 2022 **Summer School program at The Inter-University Centre for Astronomy and Astrophysics (IUCAA)**

Achievements

- 2024 Received the **Best Thesis Award in the MSc component of Integrated MSc-PhD Physics program from Hon'ble President of India Smt. Droupadi Murmu**. MSc Thesis: Disorder and Flux Induced Reentrance Topological Phase transition and Thouless pumping
- 2021 –present Receipt of **Junior Research Fellowship** of Department of Atomic Energy, Government of India, as an Integrated-Ph.D. research scholar in the National Institute of Science Education and Research Bhubaneswar
- 2018 - 2021. Receipt of **INSPIRE National Scholarship (SHE) by DST by Govt.of India**
- 2018 Ranked **9th(99.99 percentile) in West Bengal State in India on 12TH Board Exam(WBCHSE)**

- 2021 Qualified **Joint Entrance Screening Test (JEST) RANK: All India Rank 169** conducted by Indian research institutions.
- 2021 Qualified **Graduate Aptitude Test in Engineering (GATE)** jointly conducted by IITs and IISc.

Referee

Dr. Tapan Mishra

Reader-F, School of Physical Sciences
National Institute of Science Education
and Research, Bhubaneswar
✉ mishratapan@gmail.com